

IntelLogixAI

Liya Biju & Raihan Zakheer

February 18, 2025

Introduction

Introduction to IntelLogixAI

IntelLogixAI is an AI-powered Accounts Management System designed to streamline user authentication, role-based access, and provide AI-driven business insights for logistics companies. As businesses scale, managing user roles and ensuring data security becomes increasingly complex. IntelLogixAI addresses these challenges by offering advanced security features, predictive analytics for sales and performance, and seamless integration with existing systems.

The platform helps logistics companies optimize operations, make data-driven decisions, and improve security, ultimately driving growth and efficiency. With scalability and modern AI tools at its core, IntelLogixAI empowers businesses to adapt and thrive in a dynamic logistics environment.

Problem Statement

Logistics companies often face significant challenges in managing user accounts, ensuring secure access to sensitive data, and gaining actionable insights from business data. These challenges include:

Complex User Role Management: As organizations grow, manually managing various user roles and permissions becomes increasingly difficult, leading to potential security risks and inefficiencies.

Inefficient Authentication: Current authentication methods may lack sufficient security features, making sensitive business data vulnerable to unauthorized access.

Lack of Predictive Analytics: Many companies struggle to analyze historical data effectively, resulting in missed opportunities for optimizing sales, performance, and operations.

IntelLogixAI addresses these issues by providing a comprehensive, AI-powered solution that automates user management, enhances security, and offers predictive analytics to improve business decision-making.

Objective

Primary Objective

The main objective of IntelLogixAI is to enhance operational efficiency, improve system security, and provide actionable predictive insights for business growth. The system will utilize AI algorithms to analyze data, identify inefficiencies, and provide actionable recommendations.

Goals to Achieve:

- ▶ Streamline account management and minimize administrative overhead.
- ▶ Enhance security with AI-powered authentication and role-based access.
- ▶ Leverage AI for predictive analytics that provide business insights, such as forecasting sales, growth potential, and operational bottlenecks.

Purpose

Purpose of the System

The purpose of IntelLogixAI is to define and implement a robust, AI-powered Accounts Management System specifically designed for logistics operations. This system aims to automate and optimize user authentication, profile management, and AI-driven analytics to ensure smooth operations across all levels of the logistics company.

Key Features:

- ▶ Seamless user authentication process with role-based access controls.
- ▶ Comprehensive profile management system for employees and clients.
- ▶ AI-powered analytics for deeper insights and smarter decision-making.

Scope

Scope of the System: The IntelLogixAI system will be responsible for managing user accounts, handling role-based access, and providing advanced data analytics to support business decisions. The system will be designed for scalability, integrating third-party authentication services and ensuring compliance with security standards. **Core Components:**

- ▶ **User Authentication:** Multi-factor authentication, password policies, and session management.
- ▶ **Role-based Access Control (RBAC):** Different levels of access for employees, managers, and administrators.
- ▶ **AI Analytics:** Data analysis and forecasting tools to predict trends, optimize logistics, and analyze performance.
- ▶ **Third-party Integrations:** OAuth, SAML, or other industry-standard integrations for secure login and external services.

Social Relevance of the Project

The IntelLogixAI project is socially relevant as it addresses key challenges faced by the logistics industry, which plays a crucial role in the global economy. By improving operational efficiency, enhancing security, and enabling data-driven decision-making, the project supports businesses in achieving sustainable growth and competitive advantage.

- ▶ **Enhancing Job Efficiency:** Automates tasks and provides insights, allowing employees to focus on high-value work, boosting productivity.
- ▶ **Promoting Data Security:** Utilizes AI-driven security to protect sensitive data, preventing breaches and building trust.
- ▶ **Supporting Economic Growth:** Optimizes logistics and decision-making, improving supply chain efficiency in a dynamic global market.

SRS

Functional Requirements

Employee Contacts – Manage employee details, roles, and access.

Sales – Track sales, generate reports, and compare periods.

Purchase Order & Quotation – Automate PO creation and vendor comparison.

Dashboard – Display sales trends, top performers, and alerts.

History of Sales – Maintain searchable and exportable sales records.

Inventory – Monitor stock, automate restocking, and track products.

Products – Analyze sales performance and pricing trends.

Employee Profile – Manage roles, achievements, and permissions.

User Authentication – Secure login with role-based access.

AI-Driven Predictions – Forecast growth, analyze trends, and track key clients.

SRS

Non Functional Requirements

Performance – Support 100 concurrent users, 1,000 transactions/min, and fast dashboard/report loading.

Response Time – Login/profile updates within 2s, API responses within 1s for large datasets.

Scalability – Handle up to 1M users and scale horizontally for higher loads.

Security – Use bcrypt for password hashing, enforce MFA for high-privilege accounts.

Reliability – 99.9% uptime, with maintenance under 2 hours/month.

Error Handling – Log errors with details, show user-friendly messages, and trigger critical alerts.

Data Privacy – Secure PII storage, restricted access, and user rights for data export/deletion.

Technology Stack

Frontend: React.js

React.js is used for its efficient component-based architecture, allowing for dynamic updates to the UI without reloading the page. It will enable seamless navigation and responsiveness across different devices.

Backend: Node.js and Express.js

Node.js, along with Express.js, provides a lightweight and scalable solution for handling server-side operations. It is ideal for handling high-concurrency requests and managing APIs for user authentication and role management.

Database: PostgreSQL

PostgreSQL is an open-source relational database management system. It provides reliability, scalability, and advanced features like complex queries, full-text search, and robust transaction handling.

System Architecture Overview

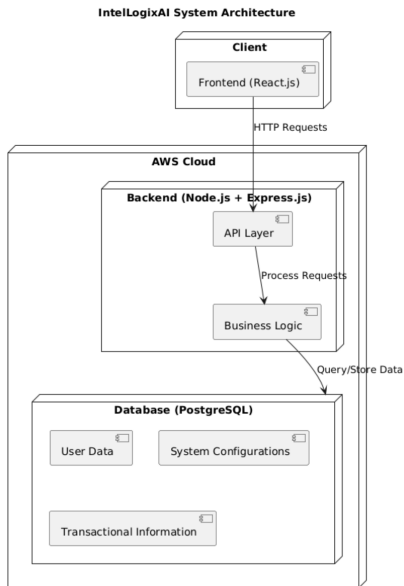


Figure: System Architecture

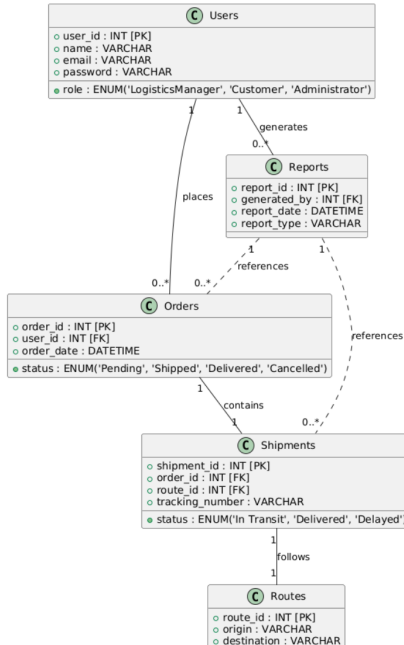


Figure: usecase

User Interaction Summary:

- ▶ **Admin:** Manages user roles and permissions, configures system settings, oversees system security.
- ▶ **Manager:** Views sales reports, tracks team performance, generates forecasts.
- ▶ **Employee:** Updates contact details, accesses personal sales data, interacts with clients.

Database Schema Overview



Development Progress

- ▶ **User Authentication Module:** 100% completed. created login and signup with otp verification using node mailer.
- ▶ **Invoice Generation:** 100% completed.
- ▶ **Home Page:** 60% completed. more modules and features has to be added
- ▶ **Contact Details:** 100% completed.

Summary

- ▶ IntelLogixAI aims to revolutionize logistics management by combining secure user account management, AI-driven insights, and robust data analytics.
- ▶ The system provides scalable, secure, and efficient solutions to manage accounts and optimize business operations.
- ▶ Future enhancements include integrating additional third-party services, expanding AI capabilities, and adding new features for improved operational insights.